JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 96, NO. B10, PAGES 16,061–16,067, SEPTEMBER 10, 1991

The Patagonian Orocline: New Paleomagnetic Data From the Andean Magmatic Arc in Tierra del Fuego, Chile

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The Hardy Formation is a 1300-m-thick succession of Upper Jurassic-Lower Cretaceous volcaniclastic sedimentary rocks interbedded with lava flows on Hoste Island at the southernmost tip of South America (55.5°S, 291.8°E). The strata are gently folded and metamorphosed to the prehnite-pumpellyite grade. A well-defined characteristic direction of magnetization, carried by magnetite, was readily identified in 95 samples from seven sites. At a given site, the directions group slightly better without structural correction. However, the means of the seven sites cluster better without tilt correction at the 99% significance level, implying that the magnetization postdates the folding event. It is most likely that the magnetization was acquired during the mid- to Late Cretaceous Andean orogeny that involved the folding and emplacement of the Patagonian Batholith. The fact that all samples are normally magnetized supports this age assignment. The pole position of 42.9°N, 156.6°E, α_{95}=3.3° implies that the sampling area has rotated counterclockwise relative to cratonic South America by 90.1 ± 11.9° with no significant flattening of inclination (F=1.9 ± 3.7°). Geologic considerations indicate that the rotation involved the entire Andean magmatic arc in Tierra Del Fuego. The results support interpretation of the Hardy Formation as part of the Andean magmatic arc deposited on the Pacific side of the Late Jurassic-Early Cretaceous Rocas Verdes marginal basin. Oroclinal bending of the arc in southernmost South America accompanied inversion of the marginal basin and the development of a Late Cretaceous-Cenozoic left-lateral transform system (South America-Antarctica) that later developed into the North Scotia Ridge.

INTRODUCTION

Between latitudes 50°S and 56°S the trend of the Andean Cordillera changes abruptly from a southerly direction to an easterly direction. This bend connects the main Andean belt in southern Chile and Argentina to the submarine ridges and emergent continental fragments of the North Scotia Ridge (Figure 1) and was named the Patagonian orocline by *Carey* [1955]. A longstanding problem that has confronted workers trying to resolve the tectonic evolution of the southernmost Andes and its relation to the Antarctic Peninsula-Scotia arc region has been determining whether the Patagonian orocline formed as a curved orogen, is a nonrotational arc that contains offset domains bounded by parallel, left-lateral strike-slip faults, or whether the bend is the product of orogenic rotation [*Marshak*, 1988].

Previous studies of bends in orogenic belts have demonstrated the utility of paleomagnetic analyses to determine the sense, magnitude, and timing of oroclinal bending [*Eldredge et al.*, 1985, and references therein; *Beck*, 1988]. These studies are based on the assumption that bending of an orogen results in progressive variations in paleomagnetic declinations around the bend. Comparison of paleomagnetic declinations from rocks collected around the bend with declinations obtained from stable cratonic rocks of the same age can provide sense and magnitude of rotation. Studies by *Dalziel et al.* [1973] and *Burns et al.* [1980] utilized paleomagnetism to address the problem of oroclinal bending in the southernmost Andes. Both studies concluded that some counterclockwise rotation of the southern tip of the South American continent has occurred since the Late Cretaceous, although the studies differed on the amount of rotation. Unfortunately, neither study presented data that pass modern confidence "filters" for data reliability [*Beck*, 1988].

In this study we present new paleomagnetic data from rocks of the Lower Cretaceous Andean magmatic arc on Peninsula Hardy in extreme southern Chile (Figure 1). Our results strongly indicate that the Patagonian orocline is the product of approximately 90° of counterclockwise crustal rotation since the mid-Cretaceous.

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Paper number 91JB01498.
0148-0227/91/91JB-01498$05.00

GENERAL GEOLOGY

Southernmost Andes

Southernmost South America can be subdivided into four broad lithotectonic provinces from southwest to northeast: (1) the Patagonian batholith, a Late Jurassic to mid-Tertiary magmatic arc province, (2) the Rocas Verdes basin, a collapsed marginal basin of Late Jurassic to Early Cretaceous age, (3) the metamorphic hinterland, a multiply deformed uplifted metamorphic belt that was formerly the continental margin of the Rocas Verdes basin, and (4) the Late Cretaceous to Tertiary foreland fold and thrust belt of the Magallanes basin foredeep (Figure 1).

With the onset of Gondwana breakup in the Middle to Late Jurassic, southernmost South America has experienced a complex sequence of Mesozoic and Cenozoic tectonic events that is the focus of ongoing study and will only be briefly summarized here. (For a more thorough review of the structural and tectonic evolution of the region, see *Dalziel* [1989].) The oldest rocks in the region consist of Paleozoic to lower Mesozoic metasediments and metavolcanics that are interpreted as an ancient accretionary prism complex formed along Gondwana's Pacific margin [*Forsythe and Mpodozis*, 1979; *Dalziel*, 1982; *Dalziel and Forsythe*, 1986]. Initial Gondwana breakup resulted in the extrusion of large volumes of Kimmeridgian to Tithonian silicic volcanics and contemporaneous intrusion of anatectic granites associated

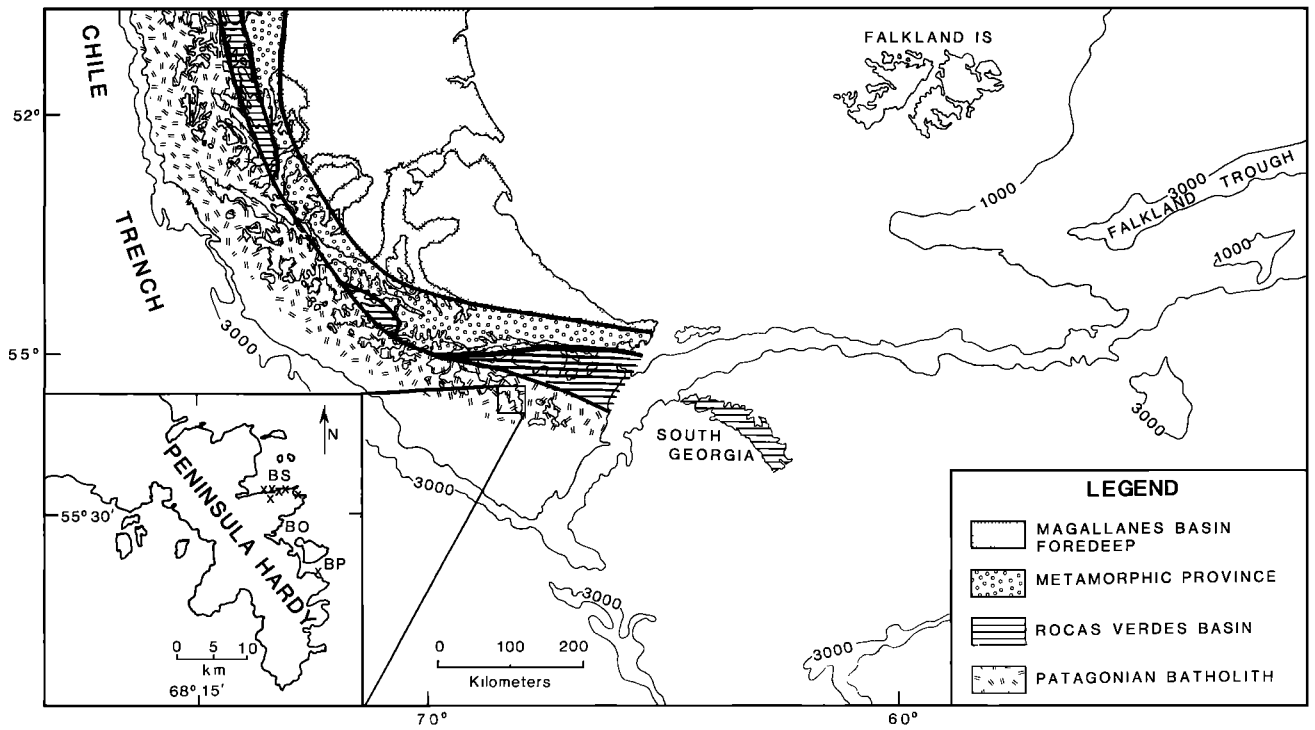


Fig. 1. Map of southernmost South America showing the four principal lithotectonic provinces and the location of Peninsula Hardy. Crosses are site locations: BS, Bahia Scotchwel; BP, Bahia Schapenham. The South Georgia platform is shown in its approximate predrift position with respect to South America [*Dalziel et al.*, 1975].

with regional extension [*Dalziel et al.*, 1974, 1987; *Mukasa et al.*, 1988]. During Late Jurassic-Early Cretaceous time the Rocas Verdes marginal basin lay inboard of a young volcanic arc [*Katz*, 1973; *Dalziel et al.*, 1974; *Bruhn and Dalziel*, 1977; *de Wit*, 1977; *Bruhn et al.*, 1978; *Dalziel*, 1981; *Mukasa et al.*, 1988]. This basin was floored in part by quasi-oceanic crust and received up to 6 km of sedimentary and volcanic infill [*Dalziel et al.*, 1974; *Dott et al.*, 1977; *Winn and Dott*, 1978]. During the mid-Cretaceous the marginal basin was closed and inverted as the arc collided or was "rewelded" to the continent. This event, the main phase of the Andean orogeny [*Dalziel and Palmer*, 1979], resulted in intense shortening, regional metamorphism, uplift of the southernmost Andes, and development of a downwarped foreland basin [*Halpern*, 1973; *Nelson et al.*, 1980; *Dalziel*, 1989]. Following marginal basin inversion, shortening propagated into the foreland resulting in the development of a foreland fold and thrust belt during Late Cretaceous to Early to mid-Tertiary time [*Winslow*, 1982]. Recent work [*Dalziel and Brown*, 1989] also suggests that during latest Cretaceous-Paleogene time, part of the metamorphic core zone of the cordillera experienced tectonic denudation. Finally, a throughgoing, strike-slip fault system that now cuts across the southernmost Andes may have initiated during the mid- to Late Cretaceous and evolved during the Tertiary, and been associated with the development of the Scotia plate and the final separation of South America from Antarctica [*Barker and Burrell*, 1977; *Winslow*, 1982].

Peninsula Hardy

Peninsula Hardy is principally composed of volcanic and intrusive igneous rocks, pyroclastic deposits, and thick turbidite sequences. Except for a series of Miocene extrusive rocks, most of the dated rocks on Peninsula Hardy are Upper Jurassic to Lower Cretaceous based on fossil evidence and are interpreted as a magmatic arc assemblage [*Suarez and Pettigrew*, 1976; *Suarez et al.*, 1985]. The Hardy Formation is the name given to a 1300 m thick succession of Upper Jurassic-Lower Cretaceous volcaniclastic rocks and interbedded lava flows found on Peninsula Hardy [*Suarez and Pettigrew*, 1976]. The rocks are gently folded and metamorphosed to zeolite or prehnite-pumpellyite grade. Both intensity of deformation and metamorphism increase northward toward the continent [*Bruhn and Dalziel*, 1977].

Data from five different sites from the south coast of Bahia Scotchwel, one from the coast southeast of Bahia Scotchwel and one from the coastline south of Bahia Schapenham (Figure 1), are presented here. Sample lithologies from these sites consist of thinly laminated mudstones, tuffaceous siltstones, graywackes, and a massive pyroclastic unit. Examination of thin sections of these units reveals that quartz, plagioclase, lithic fragments, calcite, metamorphic chlorite, fossil fragments, clays, pyroxene, and various opaques are the principal constituents for these rock types.

METHODS AND PRODEDURES

Two of us (W.D. Cunningham and K.A. Klepeis) spent 4 weeks on Peninsula Hardy in May 1988 collecting samples for a paleomagnetic study. Peninsula Hardy was chosen because of the low metamorphic grade of the fossiliferous rocks found there and because it is located on the less deformed outer arc of the Patagonian orocline oceanward of the ophiolitic complexes of the Rocas Verdes back-arc basin. A variety of rock types was sampled including fine- and coarse-grained sedimentary rocks, a mafic sill, volcaniclastic breccias, debris flow deposits, and a massive pyroclastic unit. Only those rocks that could provide an indication of paleohorizontal attitude were

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sampled. Almost all sampling sites were located along coastal areas where access and good rock exposure were available. Most paleomagnetic samples were drilled in the field, but owing to difficulties with the drilling equipment, some oriented hand samples were also taken.

The remanent magnetization was measured with a cryogenic magnetometer. All samples were progressively thermally demagnetized in five to eight steps. The field in the cooling region following thermal demagnetization was less than 3 gammas. Throughout the experimental procedure the samples were kept in a magnetically shielded room where the magnetic field was approximately 100 gammas. The data were analyzed by calculating linear trends in orthogonal vector projections. Of the 94 samples, only two samples from site 593 were rejected from the analysis because their directions differed from the site mean by more than 2 angular standard deviations.

EXPERIMENTAL RESULTS

The natural remanent magnetization (NRM) directions of all samples point westerly with steep negative inclinations. The NRM intensities are of the order of 10^{-6} to 10^{-8} A m^2/kg. Upon progressive thermal demagnetization, most samples retained 60% or more of their NRM intensity up to about 400°C. Above this temperature, the intensities of most samples decreased markedly upon approaching the magnetite Curie point (Figure 2, sample 5162-3). Some samples contain a distinctive hematite component (Figure 2, sample 593-10). The directions changed very little during demagnetization up to 570°C. At higher temperatures, the directions started to scatter. Vector component diagrams confirm the predominance of only one component of magnetization with an occasional minor overprint which is readily removed by heating to 150° or 200°C. This holds true for both the finer grained sediments sampled at sites 5161, 5162, and 5181 and the coarser grained sediments at sites 5191, 551, 593, and 5182 (Figure 3).

The directions of characteristic remanent magnetization (ChRM) were calculated with the aid of vector component diagrams, typically encompassing the temperature range from 150° to 570° or 600°C. However, because most samples contain one dominant component of magnetization, it makes little difference for the statistical analysis whether one uses the directions at a particular demagnetization step or the ChRMs. Figure 4 illustrates this point for the data from site 593. It compares the mean directions and two statistical parameters for each demagnetization step with the corresponding values obtained using the ChRM. Only minor variations are observed

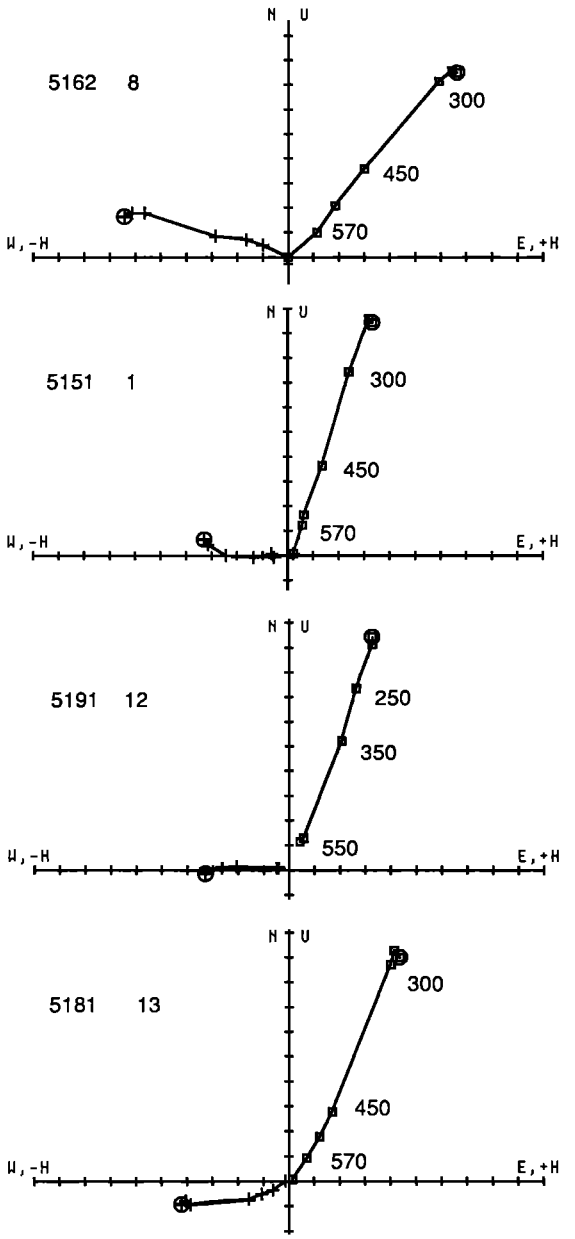


Fig. 3. Vector component diagrams. Squares are the projection onto the northeast plane; crosses are in the up-horizontal plane. Numbers indicate the demagnetization temperature in degrees Celsius. Data are uncorrected for bedding tilt.

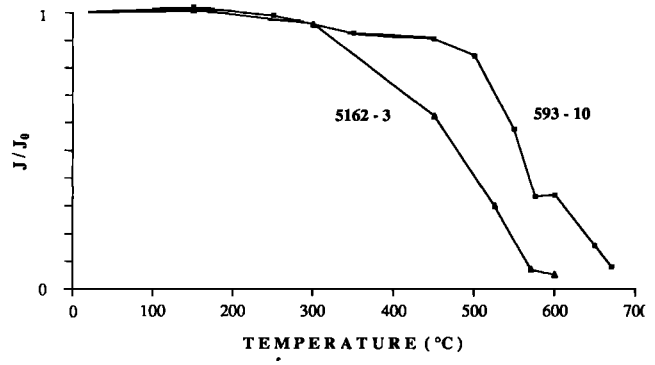


Fig. 2. Relative change in intensity during thermal demagnetization. Sample 5162-3 is dominated by magnetite, and sample 593-10 contains both magnetite and hematite.

in the temperature range from 250° to 600°C. The other sites show a similar behavior.

The site means, based on the characteristic remanent magnetization, and the statistical parameters are listed in Table 1. All sites yield good results as evidenced by the small circles of confidence (α_{95}) and the large values for Fisher's precision parameter k. For all but one site the ChRM directions cluster better without structural correction. The variations in bedding attitudes at a given site are too small to make the improvement statistically significant at the 90% confidence level with the exception of site 593 where the improvement is significant at the 95% level. The F test can also be applied to the means of the site means because the tilt corrections for the sites are different. The test shows that the in situ site means group better at the 99% significance level. Thus the observed

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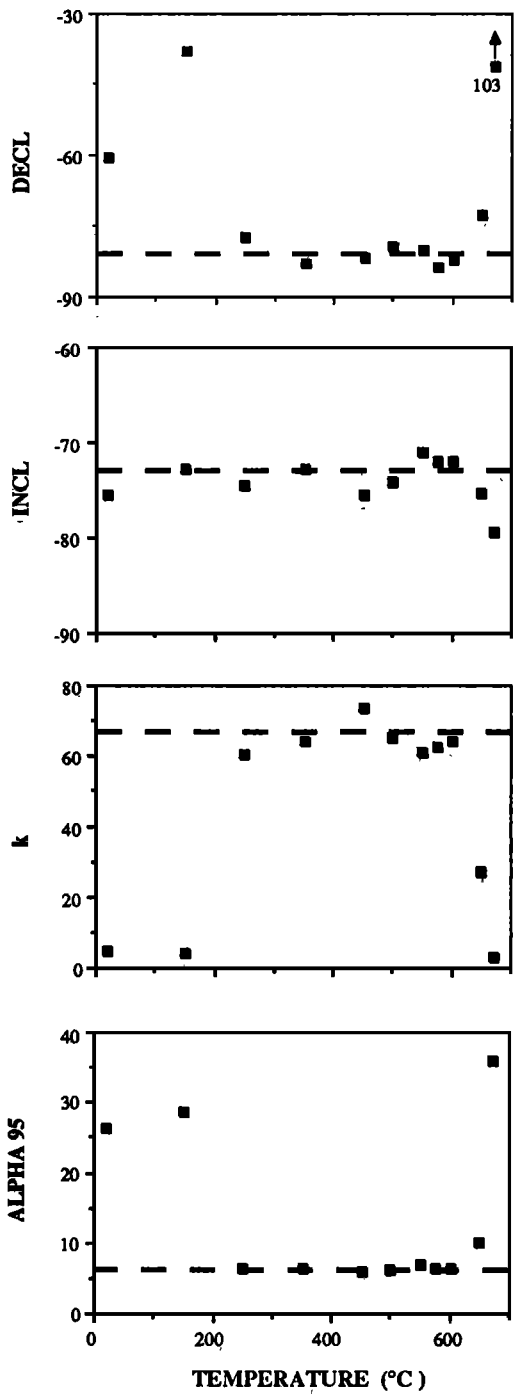


Fig. 4. Comparison of the mean direction and two statistical parameters from site 593 for each demagnetization step with the corresponding values based on the characteristic remanent magnetization (heavy dashed lines).

magnetization was acquired after the folding event. The in situ directions of the characteristic remanent magnetization of all sites are shown in Figure 5.

AGE OF MAGNETIZATION

On the basis of paleontological evidence, the age of the Hardy Formation ranges from Late Jurassic through Early Cretaceous [*Suarez and Pettigrew*, 1976], a time span characterized by frequent geomagnetic reversals [*Harland et al.*, 1982]. Because the sampling sites cover over 300 m of stratigraphic section (probably representing a large span of time), one would expect both normal and reversed polarity directions of magnetization if samples carried a primary depositional remanence. However, all samples are normally magnetized, suggesting remagnetization as was indicated by the fold test. Because of the consistent normal polarity, we consider it likely that the Hardy Formation was remagnetized during the Cretaceous normal polarity superchron (118-83 Ma). This is also the time of the Andean orogeny with associated regional metamorphism, deformation, and batholith emplacement that is interpreted to have occurred between 105 and 90 Ma [*Halpern*, 1973; *Dalziel and Palmer*, 1979; *Hervé et al.*, 1984; *Biddle et al.*, 1986; *Mukasa et al.*, 1988; *Dalziel*, 1989]. Thus we interpret the magnetization of the Hardy Formation as a thermoremanent magnetization acquired by heating and cooling during the middle part of the Cretaceous.

DISCUSSION

The mean direction of our seven sampling sites, uncorrected for bedding tilt, has a declination of 265.8°, an inclination of -72.6°, and a circle of confidence of $\alpha_{95} = 4.2°$. As a reference pole for cratonic South America we use the Cretaceous pole calculated by *Castillo et al.* [1991] which is based on 11 determinations (87.6°S, 31.0°E, $\alpha_{95} = 3.3°$). The reference pole yields an expected direction for Peninsula Hardy of D= 355.9° and I=-70.7°. Comparison of the observed and expected directions implies a counterclockwise rotation of the sampling area of R = 90.1°±11.9° using the method of *Beck* [1980] and *Demarest* [1983] (Figure 6). The flattening of the inclination which corresponds to a change in latitude is F = 1.9°±3.7° and is not statistically significant.

Although our sampling area is limited to Peninsula Hardy, we believe that the counterclockwise rotation of Peninsula Hardy was not an isolated block rotation but part of a regional rotation involving at least the magmatic arc and perhaps the entire orogen. Previous tests for oroclinal bending from other parts of the southernmost Andes [*Dalziel et al.*,1973; *Burns et al.*, 1980] do suggest, at least qualitatively, counterclockwise rotation of the southernmost Andes since the mid-Cretaceous. We believe that the overall 90° change in physiographic and structural trend in the southernmost Andes and the 90° counterclockwise rotation of Peninsula Hardy are not coincidental.

The large counterclockwise rotation suggested by the paleomagnetic data has important implications for Scotia arc evolution and Gondwana reconstructions between South America and Antarctica. Removal of the rotation from the southernmost Andes creates an approximately rectilinear trend for the magmatic arc prior to the Late Cretaceous. This disagrees with some previous workers' interpretations that there was an original cusp or indentation along the active margin in the area that later marked the zone of separation between the South America and Scotia plates [*Griffiths and Barker*, 1972]. If southernmost South America was not bent originally, then Gondwana reconstructions must allow for lateral overlap or adjacency between the northern Antarctic Peninsula and southernmost South America or somehow fit the Antarctic Peninsula so that it lines up at the southern end of the Andean Cordillera (Figure 7) [*Norton and Sclater*, 1979; *Lawver and Scotese*, 1987; *Grunow et al.*, 1987]. It is possible, however, that an originally curved magmatic arc was tectonically straightened during opening of the Rocas Verdes basin during the Middle to Late Jurassic and subsequently

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TABLE 1. Summary of Paleomagnetic Data From Peninsula Hardy (55.5°S, 291.8°E)

Site	Lithology	*N*	R	*D*, deg	*I*, deg	k	α_{95}	Latitude	Longitude
5161	Mudstone/Siltstone	18/0	17.82	246.6	-71.9	97.0	3.5	-34.5	329.4
			17.82	264.3	-67.0	93.5	3.6		
5162	Mudstone/Siltstone	12/0	11.67	290.5	-69.5	33.1	7.7	-51.1	355.2
			11.63	344.1	-70.9	30.0	8.1		
5181	Mudstone/Siltstone	13/0	12.90	251.4	-70.8	115.1	3.9	-35.0	333.1
			12.88	279.4	-63.4	99.6	4.2		
5191	Siltstone/Sandstone	14/0	13.88	271.8	-73.2	105.6	3.9	-45.7	339.4
			13.85	288.4	-56.1	86.2	4.3		
551	Sand/Silt/Mudstone	18/0	17.36	256.1	-69.1	26.7	6.8	-34.9	337.8
			17.08	257.5	-84.0	18.4	8.3		
593	Sand/Silt/Mudstone	9/2	8.88	278.9	-73.0	66.7	6.4	-48.5	342.9
			8.65	337.5	-63.1	23.1	10.9		
5182	Pyroclastic Deposit	8/0	7.74	266.3	-76.4	26.7	10.9	-46.5	331.1
			7.74	235.4	-59.8	26.7	10.9		
Mean		7/0	6.97	265.8	-72.6	212.1	4.2	-42.5	337.8
			6.72	286.5	-70.6	21.5	13.3		

N is the number of samples used/rejected; R is the resultant vector; *D* is declination; *I* is inclination; *k* is Fisher's precision parameter; α_{95} is the radius of the circle of 95% confidence; Latitude and Longitude are the latitude and east longtiude of the corresponding virtual geomagnetic pole. For each site, the data in the first line are without structural correction, the data in the second line are with structural correction applied.

oroclinally bent after magnetization during basin closure. This is in keeping with the obvious reentrant in the Pacific margin of the Gondwana craton noted by *Dalziel* [1982].

An important question that cannot be answered at this time is whether oroclinal bending was the product of marginal basin closure during the mid-Late Cretaceous Andean orogeny or of Cretaceous-Tertiary strike-slip tectonics and Scotia plate development or both. We believe that the bending probably started sometime during or after the Andean orogeny or post-100 Ma (following regional metamorphism on Peninsula Hardy and acquisition of secondary magnetization) and probably ceased prior to the development of the Scotia Plate at approximately 35 Ma. This minimum age for rotation is based on the following considerations. The northeast trending continental margin southeast of Cape Horn is today approximately parallel to the northeast trending extinct spreading center in the middle of the Scotia Sea, from which it presumably rifted [*British Antarctic Survey*, 1985]. It seems likely that if the southern Andes had continued to rotate in a counterclockwise sense while the Scotia plate spreading center was active (35 -10 Ma [*Barker and Burrell*, 1977]), then the continental margin and spreading center axis would have lost

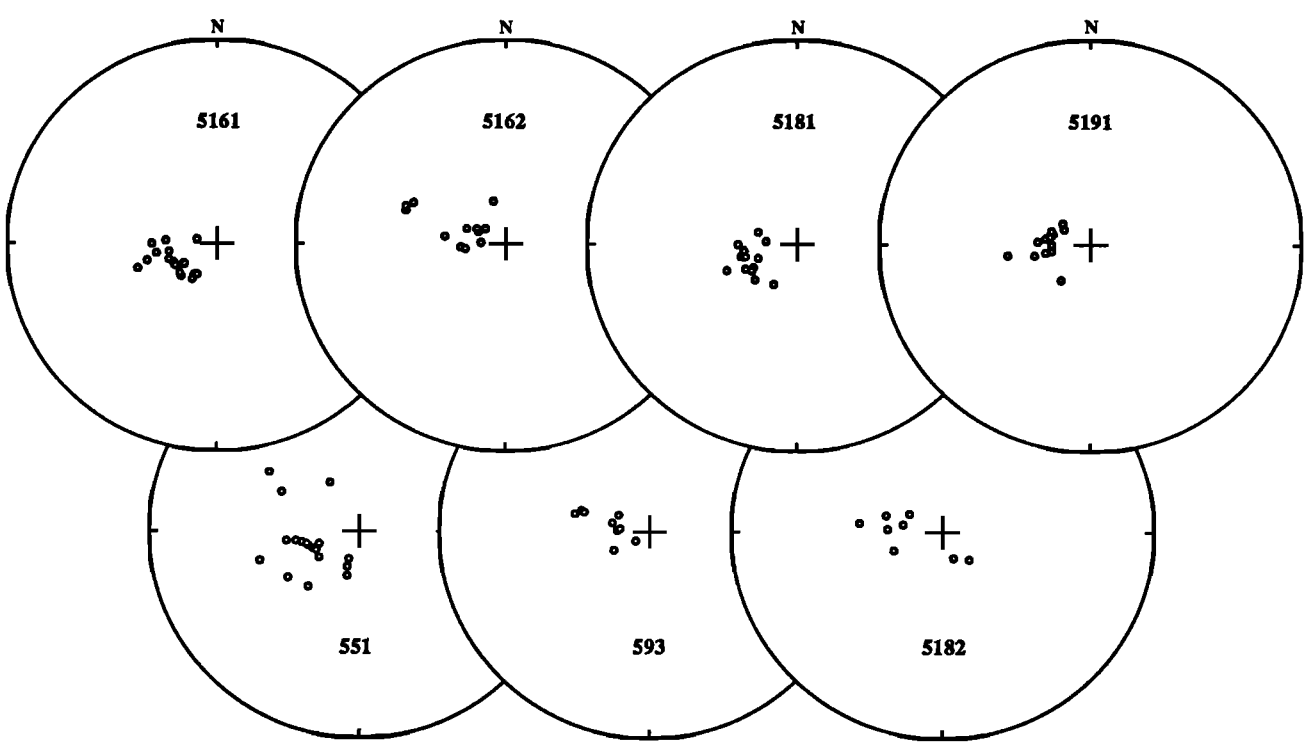


Fig. 5. Equal-area projections of the directions of the characteristic remanent magnetization without structural correction. Circles are in the upper hemisphere.

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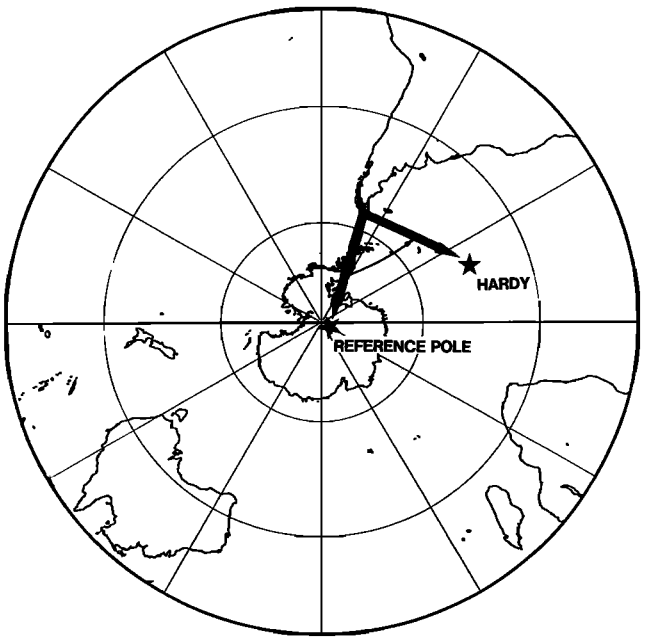


Fig. 6. Comparison of Hardy pole with mean Cretaceous reference pole for South America [*Castillo et al.*, 1991].

their parallelism. Likewise, we feel that it is unlikely that the oceanic crust of the Scotia plate was coupled to the southern tip of South America and might have rotated with it.

The opening of the South Atlantic along the South Atlantic ridge began at approximately 130 Ma and led to the westward transport of South America relative to Africa [*Lawver et al.*, 1985]. This time marked the onset of the closure of the Rocas Verdes basin and the Andean orogeny, and it is believed that Mid-Atlantic spreading resulted in forward motion of the South American plate in a mantle reference frame, as well as increased convergence rates along the Pacific margin which were accommodated by collapse of the basin [*Dalziel*, 1986]. The separation of East Antarctica from Africa along the Southwest Indian Ridge began in the Late Jurassic and led to the southwestward transport of Antarctica relative to Africa and ultimately to the southward transport of Antarctica relative to South America [*Lawver et al.*, 1985; L.A. Lawver, personal communication, 1991]. Thus since the time of the initial development of the America-Antarctic ridge, South America and Antarctica have diverged obliquely with a left-lateral component of separation (left-lateral because their westerly vector components of motion relative to Africa have different magnitudes; South America has been moving more directly westward and far faster than Antarctica's westward motion). We feel that this left-lateral oblique continental separation ultimately led to the opening of the Scotia Sea but must have been preceded by a transpressional continental wrenching system that later became transtensional before actual seafloor spreading began.

Thus we conclude that the Patagonian orocline probably formed during a time of complex regional tectonics during and/or after the Andean orogeny and preceding the opening of Drake Passage and represents a passive response to marginal basin closure and the development of a transpressional regional left-lateral shear system. It is interesting to note that similar tectonic conditions that led to the development of the Patagonian orocline may exist today east of southernmost South America where Burdwood Bank is under transpression [*Ludwig and Rabinowitz*, 1982] and the Falkland Trough is "zippering shut" from west to east. The tectonic processes occurring today in Burdwood Bank and the Falkland Trough may be a continuation of the tectonic processes that led to the closure of the Rocas Verdes basin and the counterclockwise rotation of the southernmost Andean magmatic arc.

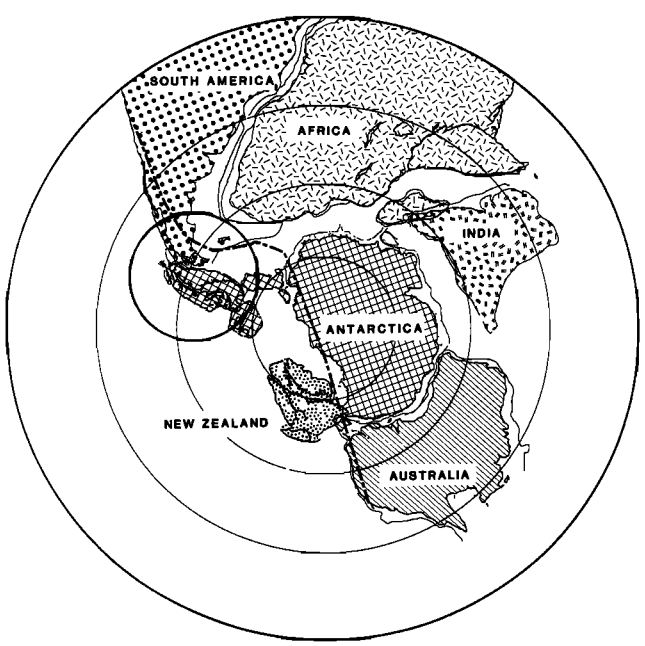


Fig. 7. Late Jurassic Gondwana reconstruction [*Grunow et al.*, 1987]. Dashed line shows approximate position of Gondwana cratonic margin. Results from this study raise questions regarding the correct fit between Antarctica and South America (circled area).

Acknowledgments. The authors thank Rich Hanson and Kip Miller for their help while in the field and the captain and crew of the R/V *Polar Duke* for transport in and out of the field. Funding was provided by the National Science Foundation Division of Polar Programs grant 86-43441 and Division of Ocean Sciences grant 87-16557. Special thanks are given to the Chilean Navy for permission to work on Peninsula Hardy and for nightly radio communication. Reviews by Robert Butler and Myrl Beck led to clarification of portions of the manuscript. This paper benefitted from discussions with Larry Lawver.

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(Received November 28, 1990;
revised May 28, 1991;
accepted June 5, 1991.)

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